Bella Melissa Ineza

Software Engineer | Application Development, APIs & Data Systems

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PROFILE

Software Engineer with 4+ years of experience developing, maintaining, and supporting production web platforms, APIs, and data-driven systems. Skilled in RESTful API development, relational database design, and application integration, with hands-on experience deploying and maintaining systems via CI/CD pipelines. Working knowledge of Java and C++ alongside strong proficiency in JavaScript/TypeScript, Python, and SQL. Comfortable troubleshooting production issues, maintaining system documentation, and collaborating across teams to keep applications reliable and performant.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Java, C++, Python, SQL, HTML, CSS,PHP

API & Integration: RESTful API design & development, API integration, third-party system integration

Databases (RDBMS): PostgreSQL, MySQL, MongoDB — schema design, data modeling, query tuning

Systems & Architecture: Authentication (JWT), Role-Based Access Control (RBAC), Object-Oriented Programming, Real-Time Data Systems

Application Support & Delivery: CI/CD (Vercel), Git, application monitoring & troubleshooting, production incident resolution, technical documentation

Tools & Platforms: Strapi, Firebase, WordPress, Cloud Deployment

PROFESSIONAL EXPERIENCE

Full-Stack Developer | Pixeleye Ltd — Kigali, Rwanda*Sept 2024 – Present*

Develop and maintain production systems and their surrounding data ecosystem, with direct responsibility for API reliability, data structure integrity, and system uptime.

- Built and maintain API-driven platforms (Next.js, Strapi, Node.js, PostgreSQL) supporting structured data management and automated content workflows — the same API/RDBMS skillset required for enterprise application support
- Migrated the Genocide Research Hub (AEGIS Trust) from a legacy content system to a modern Strapi architecture, improving scalability and maintainability — direct experience coordinating an application upgrade with minimal disruption
- Developed the HangaPitchfest platform from the ground up, enabling Rwandan startups to showcase innovations to investors — now diagnosing and resolving production issues to maintain performance and availability for this high-traffic, investor-facing system
- Delivered the RwandaFoam website (Next.js), owning frontend development and API integration end-to-end
- Maintain technical documentation for in-house developed systems and API-driven components on the 250 Podcast Lab platform
- Collaborate with cross-functional and vendor teams to deliver accessible, maintainable, and well-documented systems

Full-Stack Developer (Freelance) | Scalefix*Jul 2024 – Oct 2024*

Designed and supported a platform enabling engineering teams to monitor and coordinate scaling processes — an operational monitoring use case closely aligned with application performance tracking.

- Designed REST APIs and relational database schemas for operational data tracking
- Implemented real-time monitoring/collaboration features using Firebase
- Deployed and maintained CI/CD pipelines via Vercel to support continuous delivery and system reliability

Frontend Developer(internship) | Andela Rwanda*Aug 2023 – Apr 2024*

- Integrated backend APIs and authentication systems into production applications
- Participated in agile development cycles, code reviews, and root-cause troubleshooting
- Built reusable component systems to improve consistency and long-term maintainability

Frontend & IoT Developer(freelance) | Hub Init*Jan 2023 – Feb 2024*

Supported an IoT-enabled platform connecting physical sensor systems to cloud monitoring applications — direct experience with system health monitoring and real-time operational data.

- Designed IoT data pipelines integrating sensors with cloud platforms
- Built dashboards for visualizing and monitoring real-time operational data
- Developed backend integrations supporting device communication and data flow

SELECTED SYSTEMS & PLATFORMS

Accommodation Check-In & Revenue Monitoring Platform

- Designed a multi-tenant system with Role-Based Access Control (RBAC), authentication/authorization, and structured relational database architecture — directly relevant to enterprise user account and access management
- Built occupancy and revenue monitoring dashboards for operational oversight

Dock Tracking & Notification Platform (In Development)

- Building backend APIs and automated event detection/notification systems supporting real-time logistics monitoring — an availability and alerting use case akin to application incident detection

CURRICULUM DEVELOPMENT

Technical Curriculum Contributor — Rwanda TVET Board & International Labour Organization

- Designed structured technical training modules and aligned technical education with industry/workforce needs, working directly with government and institutional stakeholders

EDUCATION

African Leadership University — Bachelor's Degree in Software Engineering (In Progress), 2023 – Present

Rwanda Coding Academy — Diploma in Software Development & Embedded Systems, 2019 – 2022